

OpenWD workflow

1. Inputs

Model inputs

$$T_{\text{eff}}, g, \{n_i/n_{\text{ref}}\}$$

Construct the initial atmosphere

$$m, P_g(m) = g m, T(m)$$

2. Solve atmosphere

Solve the equation of state

$$\rho(m), n_e(m), \{n_s(m)\}$$

Calculate opacity and optical depth

$$\chi_\lambda \equiv \kappa_\lambda + \sigma_\lambda$$
$$\tau_\lambda(m) = \int_0^m \chi_\lambda(m') dm'$$

Solve radiative and convective transport

$$J_\lambda(m), F_{\text{rad}}(m), F_{\text{conv}}(m)$$

Flux converged?

$$F_{\text{rad}}(m) + F_{\text{conv}}(m) \simeq \sigma_{\text{SB}} T_{\text{eff}}^4$$

NO

Correct the temperature profile

$$T(m) + \delta T(m) \rightarrow T(m)$$

YES

3. Output

Calculate the emergent spectrum

$$F_\lambda = \int I_\lambda(\theta) \cos \theta d\Omega$$

Return the model

$$\{m, T, P_g, \rho, n_e\}, F_\lambda$$